Alternative Assessment Tool (AAT) Report on

“Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL”

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)
[Course Code: 23ECSPE1AD]

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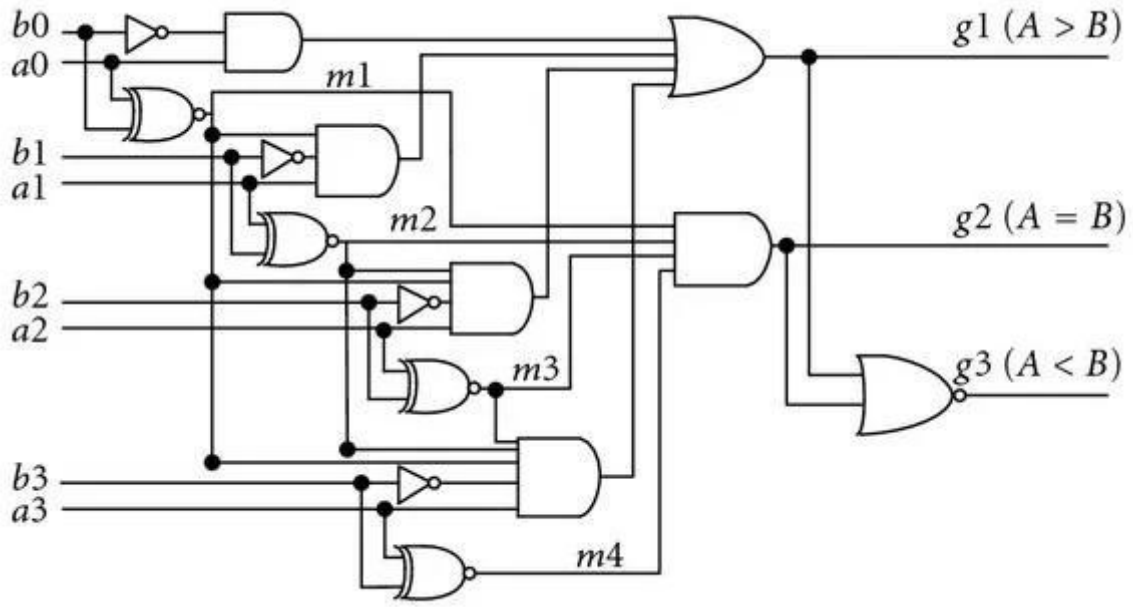
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Academic Year

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4-Bit Comparator

Circuit Diagram:



Design Code:

```
module comparator_4bit (
    input [3:0] A,
    input [3:0] B,
    output A_gt_B,
    output A_lt_B,
    output A_eq_B
);
assign A_gt_B = (A > B);
assign A_lt_B = (A < B);
assign A_eq_B = (A == B);

endmodule
```

Testbench:

```
module tb_comparator;
reg [3:0] A, B;
wire A_gt_B, A_lt_B, A_eq_B;

comparator_4bit uut (
    .A(A),
    .B(B),
    .A_gt_B(A_gt_B),
    .A_lt_B(A_lt_B),
    .A_eq_B(A_eq_B)
);

initial begin

    A = 4'b1010; B = 4'b0101;
    #10;

    A = 4'b0011; B = 4'b1001;
    #10;

    A = 4'b1100; B = 4'b1100;
    #10;

    $finish;
end

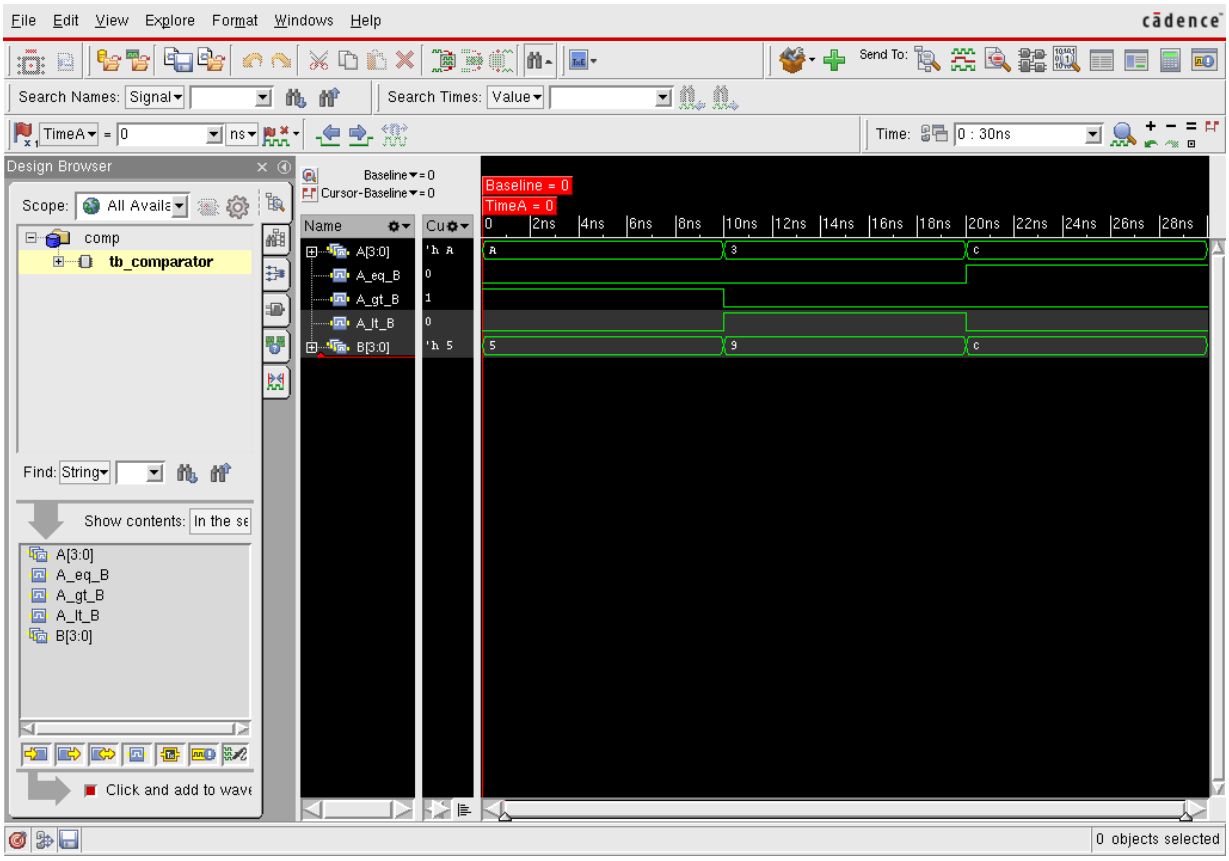
initial begin
$dumpfile("comp.vcd");
```

$dumpvars;

end

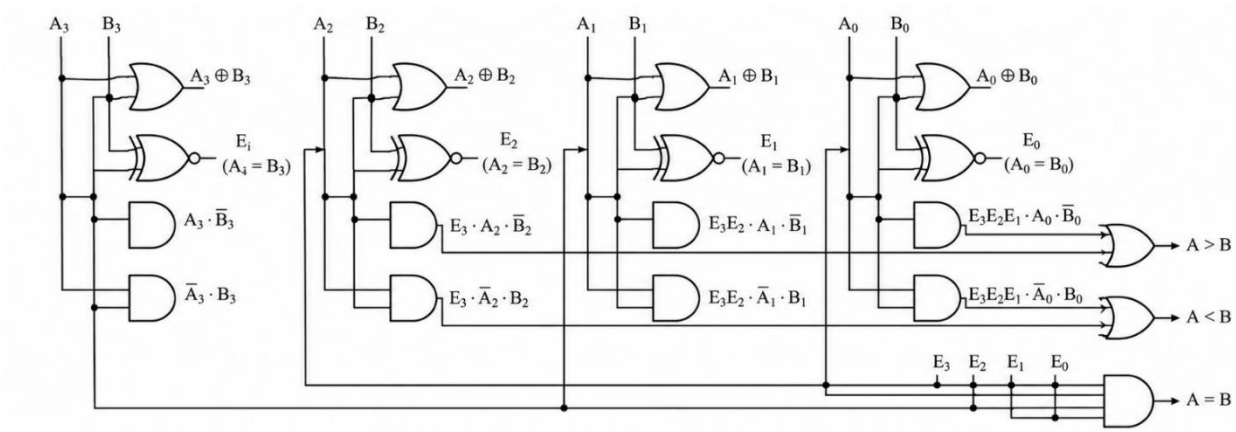
endmodule

Simulation:



4-bit Shift Register

Circuit Diagram:



Design Code:

```
module shift_register (
    input clk,
    input reset,
    input serial_in,
    output reg serial_out
);

reg [3:0] q;
always @(posedge clk or posedge reset)
begin
    if (reset)
        q <= 4'b0000;
    else
        q <= {q[2:0], serial_in};
end
always @(*)
```

```
begin
    serial_out = q[3];
end

endmodule
```

Testbench:

```
module tb_shift_register;
reg clk, reset, serial_in;
wire serial_out;

shift_register uut (
    .clk(clk),
    .reset(reset),
    .serial_in(serial_in),
    .serial_out(serial_out)
);
always #5 clk = ~clk;
initial begin
    clk = 0;
    reset = 1;
    serial_in = 0;
    #10 reset = 0;
    serial_in = 1; #10;
    serial_in = 0; #10;
    serial_in = 1; #10;
    serial_in = 1; #10;
    serial_in = 0; #10;
    #20 $finish;
```

```
end
initial begin
$dumpfile("reg.vcd");
$dumpvars;
end
endmodule
```

Simulation:

